Detailed Course Descriptions

B.Sc. (Engg.) in EEE · NSTU

Undergraduate Course Handbook

PREPARED BY

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NOAKHALI 2026

Programme Summary

Student: Md. Mohi Uddin Chowdhury

Student ID / Session: ASH2117011M

University: Noakhali Science and Technology University (NSTU), Noakhali-3814, Bangladesh

Department: Electrical and Electronic Engineering (EEE)

Degree: B.Sc. (Engg.) in EEE, 4-year Honours program, 8 semesters

Total credits: 169 (211 credit hours in total)

Credit rule: 1 credit of theory = 1 contact hour per week; lab courses = 2 contact hours per credit.

Document date: 26 October 2026

Purpose: Prepared for a Master of Science application.

This handbook-style edition preserves the course descriptions, credits and contact-hour information from the supplied Detailed Course Descriptions PDF. Its presentation follows Sample2.pdf. A dash in the source is shown as "Not specified."

Year 1 · 1st Semester

EEE 1101 — Electrical Circuits-I

Course Contents

Circuit variables and elements, Ohm's law, KCL and KVL, series-parallel circuits, source and wye-delta transformation. Nodal and mesh analysis (super node, super mesh); Thevenin, Norton, superposition, maximum power transfer and reciprocity theorems. Inductors and capacitors; AC basics (RMS value, phasors, impedance, real/reactive/apparent power, power factor improvement); natural and step response of RL and RC circuits; magnetic circuits, Faraday's law, B-H curve, hysteresis and eddy-current losses.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 1102 — Electrical Circuits-I Lab

Course Contents

Laboratory experiments verifying the theory and concepts of EEE 1101.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

CSE 1101 — Structured Programming Language

Course Contents

Problem solving and data flow diagrams; C language: data types, operators, decision making, loops, functions, recursion, pointers, arrays, strings, structures, unions, preprocessor directives, file management, bitwise operators and basic graphics programming.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

CSE 1102 — Structured Programming Language Lab

Course Contents

Programming practice in C based on the theory of CSE 1101.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

PHY 1101 — Electromagnetism, Waves, Oscillations and Optics

Course Contents

Coulomb's law, Gauss's law, electric potential, capacitors and dielectrics, magnetic field and induction, Faraday's and Lenz's laws, Maxwell's equations. Simple, damped and forced oscillations, resonance, progressive and stationary waves, Doppler effect, acoustics. Interference, diffraction, polarization, laser and nonlinear optics.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

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PHY 1102 — Physics Lab

Course Contents

Laboratory experiments based on the concepts of the physics course.

Duration: 1 Semester · **Credits:** 0.75

Contact Hours: 1.5 hours/week

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EEE 1104 — Engineering Drawing

Course Contents

Lettering, drawing instruments, sectional and isometric views of solids, plan, elevation and section of multistoried buildings, building services drawings; introduction to computer aided design (CAD) with a drawing project.

Duration: 1 Semester · **Credits:** 1

Contact Hours: 2 hours/week

MATH 1101 — Differential, Integral Calculus and Co-ordinate Geometry

Course Contents

Limits, continuity, Taylor's and Maclaurin's theorems, partial differentiation, maxima and minima, curvature. Integration methods, definite and improper integrals, Beta and Gamma functions, areas, volumes, Simpson's rule. Two- and three-dimensional co-ordinate geometry.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

ENG 1102 — Technical English Lab

Course Contents

Practice in technical English communication following the criteria of IELTS.

Duration: 1 Semester · **Credits:** 1

Contact Hours: 2 hours/week

BLWS 1101 — History of the Emergence of Independent Bangladesh

Course Contents

Geography and people of the country, partition of 1947, language movement, six-point movement, mass upsurge of 1969, election of 1970, non-cooperation movement and the War of Liberation in 1971.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

Semester Total: 20.75 credits · 26.5 contact hours/week

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Year 1 · 2nd Semester

CSE 1201 — Object Oriented Programming

Course Contents

Object-oriented concepts (abstraction, encapsulation, inheritance, polymorphism); C++ classes, constructors and destructors, friend functions, operator and function overloading, virtual functions, file streams, templates and exception handling; introduction to Java programming, inheritance and Swing events.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

CSE 1202 — Object Oriented Programming Lab

Course Contents

Programming practice in C++ and Java based on CSE 1201.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 1201 — Electrical Circuits-II

Course Contents

Series and parallel RL, RC, RLC circuits, resonance, network theorems in AC, non-sinusoidal excitation and AC transients. Poly-phase circuits (balanced and unbalanced), power calculation, magnetically coupled circuits, two-port networks and their parameters, image and characteristic impedance, passive filters (Butterworth, Chebyshev).

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 1202 — Electrical Circuits-II Lab

Course Contents

Laboratory experiments based on EEE 1201.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

PHY 1203 — Modern and Nuclear Physics, Quantum Mechanics and Thermodynamics

Course Contents

Atomic structure, crystal structure and Bragg's law, special relativity, photoelectric and Compton effects, de Broglie waves, nuclear binding energy, radioactivity, fission and fusion. Linear and angular momentum, introductory quantum mechanics (wave function, uncertainty principle, Schrödinger equation), laws of thermodynamics, Carnot cycle and entropy.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

CHE 1201 — General Chemistry

Course Contents

Chemical thermodynamics, electrochemistry (conductance, EMF, fuel and storage cells), reaction rates and kinetics, catalysis, structure and reactivity of organic molecules, instrumental methods (IR, UV, NMR, mass spectrometry), polymerization and industrial chemistry of fuels.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

CHE 1202 — General Chemistry Lab

Course Contents

Acid-base, redox and complexometric titrations, reaction kinetics, partition coefficient, viscosity, conductometric and pH-metric titrations, polymer preparation.

Duration: 1 Semester · **Credits:** 0.75

Contact Hours: 1.5 hours/week

MATH 1203 — Matrices, Ordinary and Partial Differential Equations

Course Contents

Matrices, eigenvalues and eigenvectors, vector spaces, Cayley-Hamilton theorem. Linear and exact ordinary differential equations, variation of parameters, series solutions, Bessel and Legendre equations. First- and second-order partial differential equations.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

BBA 1201 — Financial and Cost Accounting

Course Contents

Double-entry accounting cycle, financial statements and ratio analysis; cost classification, overhead, job costing, cost-volume-profit analysis, relevant cost analysis and capital budgeting.

Duration: 1 Semester · **Credits:** 2

Contact Hours: 2 hours/week

Semester Total: 20.75 credits · 24.5 contact hours/week

Year 2 · 3rd Semester

EEE 2101 — Electronics-I

Course Contents

Semiconductor and p-n junction theory, diodes (Zener, Schottky, tunnel), rectifier, clipping and clamping circuits. BJT characteristics, biasing and stability, small-signal models, audio and power amplifiers (Class A, B, AB, C). JFET and MOSFET operation, MOS amplifiers, MOSFET as a switch, CMOS inverter, differential and multistage amplifiers.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 2102 — Electronics-I Lab

Course Contents

Laboratory experiments based on EEE 2101.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 2103 — Basic Mechanical Engineering

Course Contents

Sources of energy, steam generating units, steam turbines and vapor cycles, internal combustion engines and gas turbines, refrigeration and air conditioning, fluid machinery (pumps, turbines, fans, compressors), basics of conduction, convection and radiation.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 2104 — Basic Mechanical Engineering Lab

Course Contents

Laboratory experiments based on EEE 2103.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 2105 — Electrical Machines-I

Course Contents

Electromechanical energy conversion, DC generators and motors (characteristics, starting, speed control, losses and efficiency), single- and three-phase transformers (equivalent circuit, regulation, tests, connections, vector groups, autotransformers, parallel operation) and phase conversion.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 2106 — Electrical Machines-I Lab

Course Contents

Laboratory experiments on DC machines and transformers based on EEE 2105.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 2107 — Linear Circuit Theory

Course Contents

Linearity and superposition, linear and non-linear circuits, analogous electrical, mechanical and electro-mechanical systems, Fourier transform (FFT, DFT), Laplace and Z-transforms, wavelet and Hilbert-Huang transforms, Green's function.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

MATH 2105 — Vector Analysis and Complex Variables

Course Contents

Vector algebra and calculus, gradient, divergence and curl, line, surface and volume integrals, Green's, Stokes' and Gauss's theorems. Complex functions, Cauchy-Riemann equations, Cauchy's integral formula, Laurent series, residue theorem, contour integration and conformal mapping.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

ECO 2105 — Principles of Economics

Course Contents

Basic economic concepts, national income, demand and supply, market structures, inflation, taxation, IMF and welfare economics, with reference to the economy of Bangladesh.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

Semester Total: 22.5 credits · 27 contact hours/week

Year 2 · 4th Semester

EEE 2201 — Electronics-II

Course Contents

Frequency response and Bode plots of amplifiers, feedback amplifiers, operational amplifiers (inverting, non-inverting, integrator, differentiator, summer, comparator, Schmitt trigger), 741 Op-Amp analysis, active filters, oscillators, 555 timer, VCO, PLL and blocking oscillator.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 2202 — Electronics-II Lab

Course Contents

Laboratory experiments based on EEE 2201.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 2203 — Electrical Machines-II

Course Contents

Single- and three-phase induction motors (equivalent circuit, torque-speed characteristics, starting and speed control), synchronous generators (voltage regulation, parallel operation), synchronous motors and V-curves, and special motors (universal, stepper, hysteresis, reluctance).

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 2204 — Electrical Machines-II Lab

Course Contents

Laboratory experiments on AC machines based on EEE 2203.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 2206 — Electrical and Electronic Shop Practice

Course Contents

Electrical tools, wiring, soldering, house and industrial wiring and estimation, safety devices, motor and transformer winding, testing (megger, earthing), PCB design and etching, and fault finding of electronic equipment.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 2207 — Signals and Systems

Course Contents

Classification of signals and systems, LTI systems, convolution, time-domain analysis, Fourier series and transform, Laplace transform, frequency-domain analysis, filtering, amplitude modulation and demodulation, time- and frequency-division multiplexing.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 2208 — Electrical and Electronic Circuit Simulation Lab

Course Contents

Simulation of circuits from EEE 1101, 1201, 2101 and 2201 using PSpice and MATLAB, and simulation-based design of DC and AC circuits.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

MATH 2207 — Statistics and Probability

Course Contents

Descriptive statistics, probability and conditional probability, Bayes' rule, random variables and distributions (binomial, Poisson, normal, uniform, exponential, gamma), expectation, sampling distributions, estimation, hypothesis testing and regression.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 2209 — Numerical Methods for Engineering

Course Contents

Error analysis, finite differences, interpolation, solution of nonlinear equations, linear algebraic systems (Gauss, Gauss-Seidel), curve fitting, ODEs (Euler, Runge-Kutta), numerical integration and differentiation.

Duration: 1 Semester · **Credits:** 2

Contact Hours: 2 hours/week

EEE 2210 — Numerical Methods for Engineering Lab

Course Contents

Solving mathematical problems with C/C++ programs based on EEE 2209.

Duration: 1 Semester · **Credits:** 1

Contact Hours: 2 hours/week

Semester Total: 21 credits · 28 contact hours/week

Year 3 · 5th Semester

EEE 3101 — Digital Electronics and Logic Design

Course Contents

Number systems, Boolean algebra, logic gates, combinational design and minimization, CMOS and BiCMOS gates, decoders, multiplexers, programmable logic devices, latches, flip-flops, shift registers, counters, ASM-based sequential design, wave shaping, multivibrators and timing circuits.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3102 — Digital Electronics and Logic Design Lab

Course Contents

Experiments and design of simple digital systems based on EEE 3101.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 3103 — Electrical Properties of Materials

Course Contents

Atomic models, bonding and crystal structure, electrical and thermal conduction in metals, dielectric and magnetic properties, superconductivity, band theory of solids (Bloch theorem, Kronig-Penney model), quantum mechanics basics and Maxwell-Boltzmann and Fermi-Dirac statistics.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3105 — Power System Analysis-I

Course Contents

Transmission line representation (short, medium, long), sag and stress analysis, per-unit system, symmetrical components and fault analysis, Gauss-Seidel load flow, voltage and power factor control, FACTS, HVDC and SCADA.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3106 — Power System Analysis-I Lab

Course Contents

Laboratory experiments based on EEE 3105.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 3107 — Digital Signal Processing

Course Contents

Discrete-time signals and systems, convolution, sampling theorem, Z-transform, frequency response, FIR and IIR filter structures and design, DFT and FFT algorithms, DSP processor architectures.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3108 — Digital Signal Processing Lab

Course Contents

Experiments and design of simple systems based on EEE 3107.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 3109 — Basic Communication Engineering

Course Contents

Noise and information theory, AM, FM and PM modulation, sampling, PAM, PCM, delta modulation, line coding, digital modulation (ASK, PSK, FSK), multiplexing (TDM, FDM, CDMA) and communication system design.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3110 — Basic Communication Engineering Lab

Course Contents

Laboratory experiments based on EEE 3109.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

BBA 3103 — Industrial and Business Management

Course Contents

Management functions and organization structures, personnel and financial management, facilities planning, work study, wages and incentives, production planning and control, quality engineering (ISO 9000), marketing and MIS.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

Semester Total: 24 credits · 30 contact hours/week

Year 3 · 6th Semester

EEE 3201 — Power System Analysis-II

Course Contents

Inductance and capacitance of overhead lines, ABCD line constants, insulators, underground cables, corona, swing equation and transient stability (equal area criterion), AC and DC power distribution and substation layout.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3203 — Microprocessor, Interfacing and System Design

Course Contents

16-bit microprocessor architecture, assembly programming (MASM), instruction set, I/O programming with 8255, 8251, 8253 and 8279, interrupts, data conversion algorithms, design of a microprocessor-based digital weighing machine, overview of 80286 and 80386.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3204 — Microprocessor, Interfacing and System Design Lab

Course Contents

Experiments and design of simple systems based on EEE 3203.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 3205 — Control System Engineering

Course Contents

Transient and steady-state response, Routh stability, root locus, frequency response and Nyquist criterion, block diagrams and signal flow graphs, state-variable analysis, controllability and observability, state feedback, P, PI, PD and PID controllers, introduction to PLC.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3206 — Control System Engineering Lab

Course Contents

Laboratory experiments based on EEE 3205.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 3207 — Microwave, Radar and Satellite Communication

Course Contents

Microwave links, antennas, transmission lines, waveguides, VSWR, microwave devices (Gunn, IMPATT, klystron, TWT, magnetron), satellite orbits and link design, software defined radio, multiple access techniques, radar principles and types.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3208 — Microwave, Radar and Satellite Communication Lab

Course Contents

Laboratory experiments based on EEE 3207.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 3209 — Electrical Measurement and Instrumentation

Course Contents

Static and dynamic characteristics of instruments, transducers, measurement of electrical and non-electrical quantities, oscilloscope and signal generator, acoustic instruments, noise, shielding and grounding techniques.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 3210 — Industrial Training

Course Contents

Two-week attachment to an industry or service agency after the third year to gain practical knowledge; compulsory for the degree.

Duration: 1 Semester · **Credits:** 1

Contact Hours: Not specified

EEE 3212 — Viva Voce

Course Contents

General oral examination on the basic concepts of the programme.

Duration: 1 Semester · **Credits:** 1

Contact Hours: Not specified

Semester Total: 21.5 credits · 24 contact hours/week

Year 4 · 7th Semester

EEE 4102 — Thesis/Project

Course Contents

Independent project in electrical and electronic engineering to develop initiative, creative ability and engineering judgment; results are submitted as a dissertation with drawings, charts, tables and references (continued in EEE 4202).

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4103 — Power Electronics and Industrial Drives

Course Contents

Power semiconductor devices (SCR, GTO, MOSFET, IGBT), AC-AC, DC-DC and DC-AC converters, PWM and SVM inverters, scalar and vector control of AC motors, DC motor speed control and industrial applications such as induction heating and welding.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4104 — Power Electronics and Industrial Drives Lab

Course Contents

Laboratory experiments based on EEE 4103.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4105 — Switchgear and Protection

Course Contents

Protective relaying (over-current, directional, differential, distance, microprocessor-based), fault analysis, arc interruption, AC and DC circuit breakers, fuses, bus-bar systems and reactors.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4106 — Switchgear and Protection Lab

Course Contents

Laboratory experiments based on EEE 4105.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4107 — Microcontroller and PLC based System Design

Course Contents

Microcontroller architecture (PIC16F84), programming and simulation, ADC and DAC, CISC/RISC controllers, system design examples, PLC hardware, ladder and relay logic, programming, installation and troubleshooting.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4108 — Microcontroller and PLC based System Design Lab

Course Contents

Experiments and design of simple systems based on EEE 4107.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4109 — Wireless and Mobile Communication

Course Contents

Telephony and switching systems, traffic analysis, cellular concepts, GSM and GPRS, 3G, 4G (LTE) and 5G, CDMA, OFDM and multicarrier communication, network management, ad hoc and wireless sensor networks.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4110 — Wireless and Mobile Communication Lab

Course Contents

Laboratory work based on EEE 4109.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

Semester Total: 19.5 credits · 27 contact hours/week

Year 4 · 8th Semester

EEE 4201 — Research Methodology

Course Contents

Format of paper, report and thesis writing, reference and citation styles, research analysis with simulation software, statistical analysis, questionnaire development and data survey.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4202 — Thesis/Project

Course Contents

Continuation of the thesis/project of EEE 4102, completed with a written dissertation.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4203 — Power Plant Engineering

Course Contents

Layout and principles of steam, gas turbine, combined cycle, hydro and nuclear plants, selection of plant location, load curves and load, diversity, capacity and plant factors, generation scheduling and tariff formulation.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4205 — Biomedical Instrumentation and Signal Processing

Course Contents

Biomedical transducers and signals (ECG, EEG, EMG), medical instruments, ECG and EEG signal analysis, short-time Fourier transform, filter design, noise removal, ANN-based ECG analysis, ICA, PCA, autoregressive models and FFT.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4206 — Biomedical Instrumentation and Signal Processing Lab

Course Contents

Experiments verifying the theory of EEE 4205.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4207 — VLSI and IC Fabrication Technology

Course Contents

Clean room technology, crystal growth, doping, oxidation, etching, lithography, device and IC fabrication, MOS and Bi-CMOS circuit design.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4209 — Artificial Intelligence

Course Contents

Intelligent agents, search, knowledge representation, reasoning under uncertainty, planning and learning, Prolog, neural networks, expert systems and fuzzy logic.

Duration: 1 Semester · **Credits:** 3

Contact Hours: 3 hours/week

EEE 4210 — Artificial Intelligence Lab

Course Contents

Laboratory work based on the chosen Option II course.

Duration: 1 Semester · **Credits:** 1.5

Contact Hours: 3 hours/week

EEE 4212 — Viva Voce

Course Contents

General oral examination on the basic concepts of the whole program.

Duration: 1 Semester · **Credits:** 1

Contact Hours: Not specified

Semester Total: 19 credits · 24 contact hours/week

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Option I Courses

(3 credits, 3 hours/week)

VLSI and IC Fabrication Technology

Course Contents

Clean room technology, crystal growth, doping, oxidation, etching, lithography, device and IC fabrication, MOS and Bi-CMOS circuit design.

Option II Courses

(3 credits theory and 1.5 credits lab)

Artificial Intelligence

Course Contents

Intelligent agents, search, knowledge representation, reasoning under uncertainty, planning and learning, Prolog, neural networks, expert systems and fuzzy logic.

Certification

Certified that the course descriptions above are summarised from the approved syllabus of the Department of Electrical and Electronic Engineering, Noakhali Science and Technology University.

Name and designation: Chairman / Head, Dept. of EEE, NSTU

Signature: _____

Date: _____

Official seal of the department: _____